GMAT Quantitative Reasoning

Algebra

Hard / Very Hard Problem Modeling

Challenging exercises focused on modeling complex algebraic problems. These problems require multi-step reasoning, strategic manipulation, and creative mathematical thinking. Work through the questions first, then check the detailed solutions.

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EDUCATIONAL PURPOSE: This material is intended solely for educational purposes to help students prepare for standardized tests.

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1. Part I: Exercises

Work through each problem carefully. Focus on setting up the problem correctly before attempting to solve it. These problems are designed to challenge your algebraic problem-modeling skills at a high level.

Difficulty Level: Hard to Very Hard

These problems require:

- Strategic algebraic manipulation
- Pattern recognition and identity application
- Multi-step equation solving
- Creative use of substitution and factoring

1.1. Section A: Advanced Algebraic Expressions

1.1.1. Exercise A1: Symmetric Expressions

Let $a + b = 5$ and $ab = 6$.

Questions:

1. Find the value of $a^2 + b^2$.
2. Find the value of $a^3 + b^3$.
3. Find the value of $a^4 + b^4$.
4. Express $a^5 + b^5$ in terms of $a + b$ and ab, then compute its value.

1.1.2. Exercise A2: The Difference of Powers

Questions:

1. Factor $x^4 - y^4$ completely.
2. If $x^2 + y^2 = 25$ and $x + y = 7$, find $x^4 - y^4$.
3. Prove that $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$ by expanding the right side.
4. If $a - b = 3$ and $a^2 + ab + b^2 = 19$, find $a^3 - b^3$.

1.1.3. Exercise A3: Nested Radicals and Rationalization

Questions:

1. Simplify $\sqrt{7 + 4\sqrt{3}}$ by expressing it in the form $\sqrt{a} + \sqrt{b}$.
2. Rationalize $\frac{1}{\sqrt{5}+\sqrt{3}+\sqrt{2}}$.
3. If $x = \frac{\sqrt{5}+1}{2}$ (the golden ratio), prove that $x^2 = x + 1$.
4. Find the value of $\sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + ...}}}}$ where the pattern continues infinitely.

1.1.4. Exercise A4: Polynomial Division and Remainders

Questions:

1. When $p(x) = x^3 - 2x^2 + 3x - 4$ is divided by $(x - 2)$, find the remainder using the Remainder Theorem.
2. Find the value of k such that $(x - 3)$ is a factor of $x^3 - kx^2 + 5x - 6$.
3. If $p(x)$ divided by $(x - 1)$ leaves remainder 5, and divided by $(x - 2)$ leaves remainder 8, find the remainder when $p(x)$ is divided by $(x - 1)(x - 2)$.
4. The polynomial $x^4 + ax^3 + bx^2 + cx + d$ has roots $1, 2, 3, 4$. Find $a + b + c + d$.

1.1.5. Exercise A5: Strategic Factoring

Questions:

1. Factor $x^4 + 4$ over the reals. (Hint: Add and subtract $4x^2$.)
2. Factor $x^3 + y^3 + z^3 - 3xyz$.
3. Factor $a^3 + b^3 + c^3 - 3abc$ given that $a + b + c = 0$.
4. Prove that for any integer n, the expression $n^5 - n$ is divisible by 30.

1.2. Section B: Linear Equations and Systems

1.2.1. Exercise B1: Multi-Variable Systems

Solve the following system of equations:

$$x + y + z = 6$$

$$xy + yz + zx = 11$$

$$xyz = 6$$

Questions:

1. Recognize that x, y, z are roots of a cubic polynomial. Write this polynomial.
2. Solve the cubic to find all solutions (x, y, z).
3. How many distinct ordered triples (x, y, z) satisfy the system?
4. Verify your solution by substituting back into all three equations.

1.2.2. Exercise B2: Symmetric Systems

Consider the system:

$$x + y = 10$$

$$x^2 + y^2 = 58$$

Questions:

1. Find xy using the identity $(x + y)^2 = x^2 + 2xy + y^2$.
2. Determine x and y by solving the resulting quadratic.
3. Find $x^3 + y^3$ using the identity $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$.
4. Find $(x - y)^2$ and hence $|x - y|$.

1.2.3. Exercise B3: Parameter-Dependent Systems

For what values of k does the following system have:

$$2x + 3y = 7$$

$$6x + ky = 21$$

Questions:

1. Infinitely many solutions?
2. No solution?
3. A unique solution?
4. If the unique solution satisfies $x = y$, find k.

1.2.4. Exercise B4: Diophantine Equations

Questions:

1. Find all integer solutions to $3x + 5y = 1$.
2. Find all positive integer solutions to $3x + 5y = 100$.
3. How many non-negative integer solutions exist for $x + 2y + 3z = 10$?
4. Find all pairs of positive integers (x, y) such that $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$.

1.2.5. Exercise B5: Work and Rate Problems

Questions:

1. Worker A can complete a job in 12 hours. Worker B can complete it in 18 hours. If they work together for 4 hours and then A leaves, how long does B take to finish?
2. A tank has two inlet pipes and one outlet pipe. Pipe 1 fills the tank in 10 hours, Pipe 2 in 15 hours, and Pipe 3 empties it in 20 hours. If all three operate simultaneously, how long to fill the tank?
3. Two trains travel toward each other from cities 300 km apart. Train A travels at 60 km/h and Train B at 40 km/h. A bird flies back and forth between them at 80 km/h until they meet. How far does the bird travel?
4. A boat travels 24 km upstream and 24 km downstream in a total of 10 hours. If the stream flows at 2 km/h, find the boat's speed in still water.

1.3. Section C: Quadratic Equations

1.3.1. Exercise C1: Vieta's Formulas Applications

The roots of $x^2 - 7x + 12 = 0$ are α and β.

Questions:

1. Without solving, find $\alpha + \beta$ and $\alpha\beta$.
2. Find $\alpha^2 + \beta^2$.
3. Find $\frac{1}{\alpha} + \frac{1}{\beta}$.
4. Find a quadratic equation whose roots are α^2 and β^2.

1.3.2. Exercise C2: Discriminant Analysis

For the equation $x^2 + (k-3)x + k = 0$:

Questions:

1. Find the range of k for which the equation has two distinct real roots.
2. Find the value(s) of k for which the roots are equal.
3. If one root is twice the other, find k.
4. If the roots are reciprocals of each other, find k.

1.3.3. Exercise C3: Quadratic in Disguise

Questions:

1. Solve $x^4 - 13x^2 + 36 = 0$.
2. Solve $2^{2x} - 5 \cdot 2^x + 4 = 0$.
3. Solve $x + \sqrt{x} - 6 = 0$.
4. Solve $\left(x^2 + x\right)^2 - 8\left(x^2 + x\right) + 12 = 0$.

1.3.4. Exercise C4: Optimization with Quadratics

Questions:

1. Find the minimum value of $f(x) = x^2 - 6x + 13$ and the x-value where it occurs.
2. A farmer has 100 meters of fencing. What dimensions maximize the area of a rectangular pen?
3. The sum of two positive numbers is 20. What is the maximum value of their product?
4. A ball is thrown upward with height $h(t) = -5t^2 + 30t + 10$ meters after t seconds. Find the maximum height and when it occurs.

1.3.5. Exercise C5: Integer and Rational Root Problems

Questions:

1. For what integer values of k does $x^2 + kx + 12 = 0$ have integer roots?
2. Find all rational roots of $2x^3 - 3x^2 - 11x + 6 = 0$.
3. If $x^2 + bx + c = 0$ has integer roots and $b + c = 5$, find all possible pairs (b, c).
4. The polynomial $x^3 - 6x^2 + 11x - 6$ has three integer roots. Factor it completely.

1.4. Section D: Inequalities

1.4.1. Exercise D1: Quadratic Inequalities

Questions:

1. Solve $x^2 - 5x + 6 > 0$.
2. Solve $x^2 - 4x + 4 \leq 0$.
3. Find all x satisfying $x^2 + 2x - 15 < 0$.
4. Solve $(x-1)(x-2)(x-3) > 0$ using sign analysis.

1.4.2. Exercise D2: Rational Inequalities

Questions:

1. Solve $\frac{x-2}{x+3} > 0$.
2. Solve $\frac{x^2-1}{x-2} \leq 0$.
3. Find all solutions to $\frac{1}{x} > \frac{1}{x+1}$.
4. Solve $\frac{x-1}{x+1} \geq \frac{x+1}{x-1}$.

1.4.3. Exercise D3: Absolute Value Inequalities

Questions:

1. Solve $|2x - 5| < 3$.
2. Solve $|x - 3| + |x + 2| \geq 7$.
3. Find all x satisfying $|x^2 - 4| \leq 3$.
4. Solve $|x - 1| + |x - 4| = 3$.

1.4.4. Exercise D4: Inequality Proof Techniques

Questions:

1. Prove that for all real x: $x^2 + 1 \geq 2x$.
2. Prove that for positive a, b: $\frac{a+b}{2} \geq \sqrt{ab}$ (AM-GM inequality).
3. If $a + b + c = 3$ for positive a, b, c, prove that $ab + bc + ca \leq 3$.
4. Prove that for positive x: $x + \frac{1}{x} \geq 2$.

1.4.5. Exercise D5: Systems of Inequalities

Questions:

1. Find all integer solutions to the system: $x + y < 5$ and $x - y > 1$ and $x > 0$ and $y > 0$.
2. Determine the area of the region defined by: $x \geq 0$, $y \geq 0$, $x + y \leq 4$, and $x + 2y \leq 6$.
3. Find the maximum value of $x + y$ subject to: $x \geq 0$, $y \geq 0$, $2x + y \leq 10$, and $x + 3y \leq 15$.
4. If $0 < x < 1$ and $0 < y < 1$, prove that $xy + (1 - x)(1 - y) < 1$.

1.5. Section E: Functions

1.5.1. Exercise E1: Function Composition and Inverses

Let $f(x) = 2x + 3$ and $g(x) = x^2 - 1$.

Questions:

1. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.
2. Find $f^{-1}(x)$ (the inverse of f).
3. Solve $f(g(x)) = g(f(x))$.
4. Find $(f \circ f \circ f)(x)$ and generalize to $(f \circ f \circ ... \circ f)(x)$ for n compositions.

1.5.2. Exercise E2: Domain and Range Analysis

Questions:

1. Find the domain and range of $f(x) = \sqrt{x-2} + \sqrt{6-x}$.
2. Find the domain of $g(x) = \sqrt{\frac{x-1}{x+2}}$.
3. If $f(x) = \frac{x+1}{x-1}$, find the range of f.
4. Find the domain and range of $h(x) = \frac{x}{x^2+1}$.

1.5.3. Exercise E3: Functional Equations

Questions:

1. If $f(x+1) = x^2 - 3x + 2$, find $f(x)$.
2. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x+y) = f(x) + f(y)$ for all x, y, assuming f is continuous.
3. If $f(x) + f(1-x) = 1$ for all x, find $f\left(\frac{1}{4}\right) + f\left(\frac{3}{4}\right)$.
4. If $f(xy) = f(x) + f(y)$ for positive x, y, and $f(2) = 1$, find $f(8)$.

1.5.4. Exercise E4: Piecewise and Special Functions

Questions:

1. For $f(x) = \begin{cases} x^2 \text{ if } x<0 \\ 2x \text{ if } x \geq 0 \end{cases}$, find $f(-2) + f(0) + f(3)$.
2. Find all values of x for which $\lfloor x \rfloor^2 = \lfloor x^2 \rfloor$, where $\lfloor \cdot \rfloor$ is the floor function.
3. If $f(x) = x - \lfloor x \rfloor$ (fractional part), find $f(3.7) + f(-2.3)$.
4. Solve $\lfloor 2x \rfloor = 3\lfloor x \rfloor - 1$.

1.5.5. Exercise E5: Transformations and Graphs

Questions:

1. If $f(x) = x^2$, describe the transformation that produces $g(x) = 2(x-3)^2 + 1$.
2. The graph of $y = f(x)$ passes through $(2, 5)$. What point must lie on the graph of $y = 3f(x - 1) + 2$?
3. If $f(x) = |x|$, for what value of k does $f(x-k) + f(x+k) = 2f(x)$ hold for all x?
4. Find all functions f such that $f(x) = f(-x)$ and $f(x+2) = f(x)$ for all x.

1.6. Section F: Olympiad-Style Synthesis Problems

1.6.1. Exercise F1: The Sum of Cubes

Let a, b, c be real numbers such that $a + b + c = 0$.

Questions:

1. Prove that $a^3 + b^3 + c^3 = 3abc$.
2. If additionally $a^2 + b^2 + c^2 = 6$, find $a^3 + b^3 + c^3$.
3. Find all integer solutions to $a^3 + b^3 + c^3 = 0$ with $a + b + c = 0$ and $|a| \leq 10$.
4. Prove that $(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(c + a)$.

1.6.2. Exercise F2: Telescoping and Partial Fractions

Questions:

1. Compute $\sum_{k=1}^{100} \frac{1}{k(k+1)}$.
2. Find $\sum_{k=1}^{n} \frac{1}{k(k+1)(k+2)}$.
3. Evaluate $\sum_{k=1}^{99} \frac{1}{\sqrt{k}+\sqrt{k+1}}$.
4. Compute $\prod_{k=2}^{100} \left(1 - \frac{1}{k^2}\right)$.

1.6.3. Exercise F3: Algebraic Number Theory

Questions:

1. If $x + \frac{1}{x} = 3$, find $x^3 + \frac{1}{x^3}$.
2. If $x^2 + \frac{1}{x^2} = 7$, find $x^4 + \frac{1}{x^4}$.
3. Find all real solutions to $x + \frac{1}{x} = x^2 + \frac{1}{x^2}$.
4. If $x + \frac{1}{x} = \varphi$ (golden ratio, where $\varphi = \frac{1+\sqrt{5}}{2}$), find $x^5 + \frac{1}{x^5}$.

1.6.4. Exercise F4: Competition-Style Word Problems

Questions:

1. The product of two positive integers is 2016 and their LCM is 168. Find their GCD.
2. Find all pairs of positive integers (x, y) such that $x^2 - y^2 = 91$.
3. The digits of a two-digit number sum to 9. If the digits are reversed, the new number is 27 less than the original. Find the number.
4. A three-digit number $\overline{abc}$ equals $11 \times \overline{ac}$. Find all such numbers.

1.6.5. Exercise F5: Advanced Polynomial Problems

Questions:

1. If $p(x) = x^3 + ax + b$ has a double root at $x = 2$, find a and b.
2. Find a polynomial with integer coefficients whose roots are $\sqrt{2} + \sqrt{3}$ and $\sqrt{2} - \sqrt{3}$.
3. If α is a root of $x^3 - x - 1 = 0$, express α^5 in terms of α^2, α, and constants.
4. The polynomial $p(x) = x^4 - 4x^3 + 6x^2 - 4x + 1$. Show that $p(x) = (x - 1)^4$ and find $p(2024)$.

2. Part II: Detailed Solutions

The solutions below demonstrate proper mathematical modeling using algebraic concepts. Each solution emphasizes the underlying structure and reasoning rather than trial-and-error.

2.1. Section A Solutions: Advanced Algebraic Expressions

2.1.1. Solution A1: Symmetric Expressions

Key Concept: Newton's Identities connect power sums $S_k = a^k + b^k$ to elementary symmetric polynomials $\sigma_1 = a + b$ and $\sigma_2 = ab$.

Given: $a + b = 5$ and $ab = 6$

Part 1: Find $a^2 + b^2$

We use the identity $(a + b)^2 = a^2 + 2ab + b^2$, which we rearrange to isolate $a^2 + b^2$:

$$a^2 + b^2 = (a + b)^2 - 2ab = 5^2 - 2(6) = 25 - 12 = 13$$

Answer: $a^2 + b^2 = 13$.

Part 2: Find $a^3 + b^3$

The sum of cubes identity states $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$. We already know $a + b = 5$ and $ab = 6$. From Part 1, $a^2 + b^2 = 13$, so:

$$a^2 - ab + b^2 = (a^2 + b^2) - ab = 13 - 6 = 7$$

Therefore:

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2) = 5 \times 7 = 35$$

Answer: $a^3 + b^3 = 35$.

Part 3: Find $a^4 + b^4$

We use the identity $(a^2 + b^2)^2 = a^4 + 2a^2b^2 + b^4$:

$$a^4 + b^4 = (a^2 + b^2)^2 - 2(ab)^2 = 13^2 - 2(6)^2 = 169 - 72 = 97$$

Answer: $a^4 + b^4 = 97$.

Part 4: Find $a^5 + b^5$

For higher powers, we use the recurrence relation. Multiplying $(a^4 + b^4)$ by $(a + b)$:

$$(a + b)(a^4 + b^4) = a^5 + ab^4 + a^4b + b^5 = (a^5 + b^5) + ab(a^3 + b^3)$$

Rearranging:

$$a^5 + b^5 = (a + b)(a^4 + b^4) - ab(a^3 + b^3) = 5(97) - 6(35) = 485 - 210 = 275$$

Answer: $a^5 + b^5 = 275$.

2.1.2. Solution A2: The Difference of Powers

Key Concepts:
- $x^2 - y^2 = (x+y)(x-y)$
- $x^3 - y^3 = (x-y)(x^2+xy+y^2)$
- Apply these recursively for higher powers.

Part 1: Factor $x^4 - y^4$ completely

We apply the difference of squares twice:

$$x^4 - y^4 = (x^2)^2 - (y^2)^2 = (x^2+y^2)(x^2-y^2) = (x^2+y^2)(x+y)(x-y)$$

Answer: $x^4 - y^4 = (x^2+y^2)(x+y)(x-y)$.

Part 2: Find $x^4 - y^4$ given $x^2 + y^2 = 25$ and $x + y = 7$

Using our factorization, we need to find $x - y$. From the identity $(x+y)^2 = x^2 + 2xy + y^2$:

$$49 = 25 + 2xy \Rightarrow xy = 12$$

Now $(x-y)^2 = x^2 - 2xy + y^2 = 25 - 24 = 1$, so $x - y = \pm 1$.

Therefore:

$$x^4 - y^4 = (x^2+y^2)(x+y)(x-y) = 25 \times 7 \times (\pm 1) = \pm 175$$

Answer: $x^4 - y^4 = \pm 175$.

Part 3: Prove $x^3 - y^3 = (x-y)(x^2+xy+y^2)$

Expanding the right side:

$$\begin{aligned}(x-y)(x^2+xy+y^2) &= x(x^2+xy+y^2) - y(x^2+xy+y^2) \\ &= x^3 + x^2y + xy^2 - x^2y - xy^2 - y^3 \\ &= x^3 - y^3 \checkmark\end{aligned}$$

Result: Identity verified. ∎

Part 4: Find $a^3 - b^3$ given $a - b = 3$ and $a^2 + ab + b^2 = 19$

Direct substitution into the identity:

$$a^3 - b^3 = (a-b)(a^2+ab+b^2) = 3 \times 19 = 57$$

Answer: $a^3 - b^3 = 57$.

2.1.3. Solution A3: Nested Radicals and Rationalization

Key Technique: For nested radicals of the form $\sqrt{a+b\sqrt{c}}$, try the ansatz $\sqrt{x}+\sqrt{y}$ and match coefficients.

Part 1: Simplify $\sqrt{7+4\sqrt{3}}$

Assume $\sqrt{7+4\sqrt{3}}=\sqrt{a}+\sqrt{b}$ for some positive a, b.

Squaring both sides: $7+4\sqrt{3}=a+b+2\sqrt{ab}$

Matching rational and irrational parts:
- $a+b=7$
- $2\sqrt{ab}=4\sqrt{3}$, so $ab=12$

We need two numbers with sum 7 and product 12. These are roots of $t^2-7t+12=0$, giving $t=3$ or $t=4$.

Thus $a=4$, $b=3$ (or vice versa), and:

$$\sqrt{7+4\sqrt{3}}=\sqrt{4}+\sqrt{3}=2+\sqrt{3}$$

Answer: $\sqrt{7+4\sqrt{3}}=2+\sqrt{3}$.

Part 2: Rationalize $\frac{1}{\sqrt{5}+\sqrt{3}+\sqrt{2}}$

Group and rationalize step by step. Let $S=\sqrt{5}+\sqrt{3}+\sqrt{2}$.

Multiply by $\frac{\sqrt{5}-\left(\sqrt{3}+\sqrt{2}\right)}{\sqrt{5}-\left(\sqrt{3}+\sqrt{2}\right)}$:

$$\frac{1}{S}\times\frac{\sqrt{5}-\sqrt{3}-\sqrt{2}}{\sqrt{5}-\sqrt{3}-\sqrt{2}}=\frac{\sqrt{5}-\sqrt{3}-\sqrt{2}}{5-\left(\sqrt{3}+\sqrt{2}\right)^2}$$

The denominator: $\left(\sqrt{3}+\sqrt{2}\right)^2=3+2\sqrt{6}+2=5+2\sqrt{6}$

So: $5-\left(5+2\sqrt{6}\right)=-2\sqrt{6}$

$$=\frac{\sqrt{5}-\sqrt{3}-\sqrt{2}}{-2\sqrt{6}}=\frac{\sqrt{3}+\sqrt{2}-\sqrt{5}}{2\sqrt{6}}$$

Rationalizing further by multiplying by $\frac{\sqrt{6}}{\sqrt{6}}$:

$$=\frac{\sqrt{6}\left(\sqrt{3}+\sqrt{2}-\sqrt{5}\right)}{2\times 6}=\frac{\sqrt{18}+\sqrt{12}-\sqrt{30}}{12}=\frac{3\sqrt{2}+2\sqrt{3}-\sqrt{30}}{12}$$

Answer: $\frac{3\sqrt{2}+2\sqrt{3}-\sqrt{30}}{12}$.

Part 3: Prove $x^2=x+1$ for the golden ratio

Given $x=\frac{\sqrt{5}+1}{2}$:

$$x^2 = \left(\frac{\sqrt{5}+1}{2}\right)^2 = \frac{5+2\sqrt{5}+1}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2}$$

$$x+1 = \frac{\sqrt{5}+1}{2} + 1 = \frac{\sqrt{5}+1+2}{2} = \frac{\sqrt{5}+3}{2} = \frac{3+\sqrt{5}}{2}$$

Result: $x^2 = x+1 = \frac{3+\sqrt{5}}{2}$. ∎

Part 4: Infinite nested radical

Let $L = \sqrt{2+\sqrt{2+\sqrt{2+...}}}$

Since the pattern repeats infinitely: $L = \sqrt{2+L}$

Squaring: $L^2 = 2+L$, so $L^2 - L - 2 = 0$

Factoring: $(L-2)(L+1) = 0$

Since $L > 0$, we have $L = 2$.

Answer: The infinite nested radical equals 2.

2.1.4. Solution A4: Polynomial Division and Remainders

Key Theorems:
- **Remainder Theorem:** When $p(x)$ is divided by $(x-c)$, the remainder is $p(c)$.
- **Factor Theorem:** $(x-c)$ is a factor of $p(x)$ iff $p(c)=0$.

Part 1: Remainder when $p(x)=x^3-2x^2+3x-4$ is divided by $(x-2)$

By the Remainder Theorem, the remainder is $p(2)$:

$$p(2)=2^3-2(2)^2+3(2)-4=8-8+6-4=2$$

Answer: Remainder is 2.

Part 2: Find k such that $(x-3)$ is a factor of x^3-kx^2+5x-6

By the Factor Theorem, we need $p(3)=0$:

$$3^3-k(3)^2+5(3)-6=0$$

$$27-9k+15-6=0$$

$$36-9k=0$$

$$k=4$$

Answer: $k=4$.

Part 3: Remainder when $p(x)$ is divided by $(x-1)(x-2)$

When dividing by a quadratic, the remainder has the form $r(x)=ax+b$.

Given: $p(1)=5$ and $p(2)=8$. Since $p(x)=q(x)(x-1)(x-2)+(ax+b)$:
- $p(1)=a(1)+b=a+b=5$
- $p(2)=a(2)+b=2a+b=8$

Subtracting: $a=3$, then $b=2$.

Answer: Remainder is $3x+2$.

Part 4: Find $a+b+c+d$ for $x^4+ax^3+bx^2+cx+d$ with roots $1,2,3,4$

By Vieta's formulas for a monic polynomial with roots r_1,r_2,r_3,r_4:

$$p(x)=(x-1)(x-2)(x-3)(x-4)$$

Expanding: First $(x-1)(x-4)=x^2-5x+4$ and $(x-2)(x-3)=x^2-5x+6$

$$p(x)=(x^2-5x+4)(x^2-5x+6)$$

Let $u=x^2-5x$:

$$p(x)=(u+4)(u+6)=u^2+10u+24=\left(x^2-5x\right)^2+10(x^2-5x)+24$$

$$= x^4 - 10x^3 + 25x^2 + 10x^2 - 50x + 24 = x^4 - 10x^3 + 35x^2 - 50x + 24$$

So $a = -10$, $b = 35$, $c = -50$, $d = 24$.

$$a + b + c + d = -10 + 35 - 50 + 24 = -1$$

Answer: $a + b + c + d = -1$.

2.1.5. Solution A5: Strategic Factoring

Key Technique: Sophie Germain identity and sum of cubes with three terms often appear in competition mathematics.

Part 1: Factor $x^4 + 4$ (Sophie Germain Identity)

Add and subtract $4x^2$:

$$x^4 + 4 = x^4 + 4x^2 + 4 - 4x^2 = (x^2 + 2)^2 - (2x)^2$$

Apply difference of squares:

$$= (x^2 + 2 - 2x)(x^2 + 2 + 2x) = (x^2 - 2x + 2)(x^2 + 2x + 2)$$

Answer: $x^4 + 4 = (x^2 - 2x + 2)(x^2 + 2x + 2)$.

Part 2: Factor $x^3 + y^3 + z^3 - 3xyz$

This is the famous factorization:

$$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$$

The second factor can also be written as $\frac{1}{2}[(x - y)^2 + (y - z)^2 + (z - x)^2]$.

Answer: $(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$.

Part 3: Simplify when $a + b + c = 0$

From Part 2, if $a + b + c = 0$:

$$a^3 + b^3 + c^3 - 3abc = 0 \times (a^2 + b^2 + c^2 - ab - bc - ca) = 0$$

Therefore $a^3 + b^3 + c^3 = 3abc$.

Answer: When $a + b + c = 0$, we have $a^3 + b^3 + c^3 = 3abc$.

Part 4: Prove $n^5 - n$ is divisible by 30

We need to show $30 \mid n^5 - n$, i.e., divisibility by 2, 3, and 5.

Factor: $n^5 - n = n(n^4 - 1) = n(n^2 - 1)(n^2 + 1) = n(n - 1)(n + 1)(n^2 + 1)$

Divisibility by 2: Among $n - 1$, n, $n + 1$ (three consecutive integers), at least one is even.

Divisibility by 3: Among three consecutive integers, exactly one is divisible by 3.

Divisibility by 5: By Fermat's Little Theorem, $n^5 \equiv n \pmod 5$ for all n, so $5 \mid n^5 - n$.

Since $\gcd(2, 3, 5) = 1$ and all three divide $n^5 - n$, we have $30 \mid n^5 - n$.

Answer: $n^5 - n$ is always divisible by 30. ∎

2.2. Section B Solutions: Linear Equations and Systems

2.2.1. Solution B1: Multi-Variable Systems

Key Insight: The given conditions are exactly the elementary symmetric polynomials for a cubic equation. If x, y, z are roots of $t^3 - pt^2 + qt - r = 0$, then $x + y + z = p$, $xy + yz + zx = q$, and $xyz = r$.

Given: $x + y + z = 6$, $xy + yz + zx = 11$, $xyz = 6$

Part 1: Write the cubic polynomial

By Vieta's formulas, x, y, z are roots of:

$$t^3 - 6t^2 + 11t - 6 = 0$$

Answer: $t^3 - 6t^2 + 11t - 6 = 0$.

Part 2: Solve the cubic

We look for rational roots using the Rational Root Theorem. Candidates: $\pm 1, \pm 2, \pm 3, \pm 6$.

Testing $t = 1$: $1 - 6 + 11 - 6 = 0$ ✓

So $(t - 1)$ is a factor. Dividing:

$$t^3 - 6t^2 + 11t - 6 = (t - 1)(t^2 - 5t + 6) = (t - 1)(t - 2)(t - 3)$$

Answer: The solutions are $x, y, z \in \{1, 2, 3\}$.

Part 3: Count ordered triples

The three distinct values $1, 2, 3$ can be assigned to (x, y, z) in $3! = 6$ ways.

Answer: There are 6 distinct ordered triples.

Part 4: Verification

For $(x, y, z) = (1, 2, 3)$:
- $x + y + z = 1 + 2 + 3 = 6$ ✓
- $xy + yz + zx = 2 + 6 + 3 = 11$ ✓
- $xyz = 1 \times 2 \times 3 = 6$ ✓

2.2.2. Solution B2: Symmetric Systems

Key Technique: Use symmetric function identities to find xy, then recognize x and y as roots of a quadratic.

Given: $x + y = 10$ and $x^2 + y^2 = 58$

Part 1: Find xy

From $(x + y)^2 = x^2 + 2xy + y^2$:

$$100 = 58 + 2xy$$

$$xy = 21$$

Answer: $xy = 21$.

Part 2: Find x and y

Now x and y are roots of $t^2 - 10t + 21 = 0$.

Using the quadratic formula or factoring: $(t - 3)(t - 7) = 0$

Answer: $x = 3, y = 7$ (or vice versa).

Part 3: Find $x^3 + y^3$

Using $x^3 + y^3 = (x + y)(x^2 - xy + y^2) = (x + y)((x^2 + y^2) - xy)$:

$$x^3 + y^3 = 10(58 - 21) = 10 \times 37 = 370$$

Answer: $x^3 + y^3 = 370$.

Part 4: Find $(x - y)^2$ and $|x - y|$

$$(x - y)^2 = x^2 - 2xy + y^2 = 58 - 42 = 16$$

$$|x - y| = 4$$

Answer: $(x - y)^2 = 16$ and $|x - y| = 4$.

2.2.3. Solution B3: Parameter-Dependent Systems

Key Concept: For a system $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$:
- **Unique solution:** $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
- **No solution:** $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ (parallel lines)
- **Infinitely many:** $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ (same line)

Given: $2x + 3y = 7$ and $6x + ky = 21$

Part 1: Infinitely many solutions

We need $\frac{2}{6} = \frac{3}{k} = \frac{7}{21}$, i.e., $\frac{1}{3} = \frac{3}{k} = \frac{1}{3}$.

From $\frac{3}{k} = \frac{1}{3}$: $k = 9$.

Answer: $k = 9$ gives infinitely many solutions.

Part 2: No solution

We need $\frac{2}{6} = \frac{3}{k} \neq \frac{7}{21}$.

But $\frac{2}{6} = \frac{1}{3} = \frac{7}{21}$, so the condition $\frac{3}{k} = \frac{1}{3}$ but $\frac{7}{21} \neq \frac{1}{3}$ is impossible.

If $\frac{3}{k} = \frac{1}{3}$ (i.e., $k = 9$), then all ratios are equal, giving infinitely many solutions, not none.

For no solution, we need $\frac{2}{6} = \frac{3}{k}$ but the third ratio different. However, $\frac{7}{21} = \frac{1}{3} = \frac{2}{6}$, so no value of k gives parallel (non-coincident) lines.

Answer: No value of k gives no solution.

Part 3: Unique solution

We need $\frac{2}{6} \neq \frac{3}{k}$, i.e., $\frac{1}{3} \neq \frac{3}{k}$, i.e., $k \neq 9$.

Answer: $k \neq 9$ gives a unique solution.

Part 4: Unique solution with $x = y$

If $x = y$, from the first equation: $2x + 3x = 7$, so $x = \frac{7}{5}$.

Substituting into the second: $6\left(\frac{7}{5}\right) + k\left(\frac{7}{5}\right) = 21$

$$\left(\frac{7}{5}\right)(6 + k) = 21$$

$$6 + k = 15$$

$$k = 9$$

But wait — $k = 9$ gives infinitely many solutions! Let's verify: when $k = 9$, the second equation becomes $6x + 9y = 21$, which simplifies to $2x + 3y = 7$ (same as the first). So any point on this line is a solution, including $\left(\frac{7}{5}, \frac{7}{5}\right)$.

Answer: $k = 9$, but this gives infinitely many solutions, one of which is $x = y = \frac{7}{5}$.

2.2.4. Solution B4: Diophantine Equations

Key Technique: For $ax + by = c$, find one particular solution, then the general solution is $x = x_0 + \left(\frac{b}{d}\right)t$, $y = y_0 - \left(\frac{a}{d}\right)t$ where $d = \gcd(a, b)$.

Part 1: All integer solutions to $3x + 5y = 1$

One particular solution: $x_0 = 2$, $y_0 = -1$ (since $3(2) + 5(-1) = 1$).

General solution: $x = 2 + 5t$, $y = -1 - 3t$ for any integer t.

Answer: $(x, y) = (2 + 5t, -1 - 3t)$ for $t \in \mathbb{Z}$.

Part 2: Positive integer solutions to $3x + 5y = 100$

First, find a particular solution. From $3x + 5y = 100$:

Try $y = 2$: $3x = 90$, $x = 30$. So $(30, 2)$ is a solution.

General solution: $x = 30 + 5t$, $y = 2 - 3t$.

For positive integers: $x > 0$ and $y > 0$
- $30 + 5t > 0 \to t > -6$
- $2 - 3t > 0 \to t < \frac{2}{3}$, so $t \leq 0$

So $t \in \{-5, -4, -3, -2, -1, 0\}$.

Solutions: $(5, 17), (10, 14), (15, 11), (20, 8), (25, 5), (30, 2)$.

Answer: 6 positive integer solutions.

Part 3: Non-negative solutions to $x + 2y + 3z = 10$

We count systematically. For each value of z:

$z = 0$: $x + 2y = 10 \to y \in \{0, 1, 2, 3, 4, 5\} \to$ 6 solutions $z = 1$: $x + 2y = 7 \to y \in \{0, 1, 2, 3\} \to$ 4 solutions $z = 2$: $x + 2y = 4 \to y \in \{0, 1, 2\} \to$ 3 solutions $z = 3$: $x + 2y = 1 \to y = 0$ only $\to$ 1 solution

Total: $6 + 4 + 3 + 1 = 14$.

Answer: 14 non-negative integer solutions.

Part 4: Positive integers with $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$

Multiply by $6xy$: $6y + 6x = xy$

Rearrange: $xy - 6x - 6y = 0$

Add 36: $xy - 6x - 6y + 36 = 36$

Factor: $(x - 6)(y - 6) = 36$

For positive x, y: we need $(x - 6)(y - 6) = 36$ with $x > 0$, $y > 0$.

Factor pairs of 36: $(1, 36), (2, 18), (3, 12), (4, 9), (6, 6), (-1, -36), ...$

Valid pairs (giving $x, y > 0$): $(7, 42), (8, 24), (9, 18), (10, 15), (12, 12)$ and their reverses.

Answer: $(x, y) \in \{(7, 42), (8, 24), (9, 18), (10, 15), (12, 12), (42, 7), (24, 8), (18, 9), (15, 10)\}$.

2.2.5. Solution B5: Work and Rate Problems

Key Formula: If a worker completes a job in t hours, their rate is $\frac{1}{t}$ of the job per hour. Combined rates add.

Part 1: Workers A and B

Rate of A: $\frac{1}{12}$ job/hour. Rate of B: $\frac{1}{18}$ job/hour.

Together for 4 hours: $4\left(\frac{1}{12}+\frac{1}{18}\right)=4\left(\frac{3}{36}+\frac{2}{36}\right)=4\left(\frac{5}{36}\right)=\frac{20}{36}=\frac{5}{9}$ of the job.

Remaining: $1-\frac{5}{9}=\frac{4}{9}$ of the job.

Time for B to finish: $\left(\frac{4}{9}\right)\div\left(\frac{1}{18}\right)=\left(\frac{4}{9}\right)\times 18=8$ hours.

Answer: B takes 8 hours to finish.

Part 2: Three pipes

Net rate: $\frac{1}{10}+\frac{1}{15}-\frac{1}{20}=\frac{6}{60}+\frac{4}{60}-\frac{3}{60}=\frac{7}{60}$ tank/hour.

Time to fill: $\frac{60}{7}\approx 8.57$ hours.

Answer: $\frac{60}{7}$ hours (approximately 8 hours 34 minutes).

Part 3: The bird problem

Time for trains to meet: $\frac{300}{60+40}=\frac{300}{100}=3$ hours.

The bird flies continuously for 3 hours at 80 km/h.

Answer: The bird travels $80\times 3=240$ km.

Part 4: Boat in stream

Let boat speed in still water = v km/h.

Upstream speed: $v-2$. Downstream speed: $v+2$.

Total time: $\frac{24}{v-2}+\frac{24}{v+2}=10$

$$24(v+2)+24(v-2)=10(v-2)(v+2)$$

$$48v=10(v^2-4)$$

$$48v=10v^2-40$$

$$10v^2-48v-40=0$$

$$5v^2-24v-20=0$$

Using the quadratic formula: $v=\frac{24\pm\sqrt{576+400}}{10}=\frac{24\pm\sqrt{976}}{10}$

$\sqrt{976}\approx 31.24$, so $v\approx\frac{24+31.24}{10}\approx 5.52$ km/h.

More precisely: $v=\frac{24+4\sqrt{61}}{10}=\frac{12+2\sqrt{61}}{5}$ km/h.

Answer: Boat speed $= \frac{12+2\sqrt{61}}{5} \approx 5.52$ km/h.

2.3. Section C Solutions: Quadratic Equations

2.3.1. Solution C1: Vieta's Formulas Applications

Vieta's Formulas: For $x^2 + bx + c = 0$ with roots α, β:
- $\alpha + \beta = -b$
- $\alpha\beta = c$

Given: $x^2 - 7x + 12 = 0$ with roots α, β

Part 1: Find $\alpha + \beta$ and $\alpha\beta$

By Vieta's formulas: $\alpha + \beta = 7$ and $\alpha\beta = 12$.

Answer: $\alpha + \beta = 7$, $\alpha\beta = 12$.

Part 2: Find $\alpha^2 + \beta^2$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 49 - 24 = 25$$

Answer: $\alpha^2 + \beta^2 = 25$.

Part 3: Find $\frac{1}{\alpha} + \frac{1}{\beta}$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{7}{12}$$

Answer: $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{7}{12}$.

Part 4: Quadratic with roots α^2, β^2

Sum: $\alpha^2 + \beta^2 = 25$. Product: $\alpha^2\beta^2 = (\alpha\beta)^2 = 144$.

The equation is $x^2 - 25x + 144 = 0$.

Answer: $x^2 - 25x + 144 = 0$.

2.3.2. Solution C2: Discriminant Analysis

Discriminant: For $ax^2 + bx + c = 0$: $\Delta = b^2 - 4ac$
- $\Delta > 0$: Two distinct real roots
- $\Delta = 0$: One repeated root
- $\Delta < 0$: No real roots

Given: $x^2 + (k-3)x + k = 0$

Here $a = 1$, $b = k - 3$, $c = k$.

$\Delta = (k-3)^2 - 4(1)(k) = k^2 - 6k + 9 - 4k = k^2 - 10k + 9 = (k-1)(k-9)$

Part 1: Two distinct real roots

Need $\Delta > 0$: $(k-1)(k-9) > 0$

This holds when $k < 1$ or $k > 9$.

Answer: $k < 1$ or $k > 9$.

Part 2: Equal roots

Need $\Delta = 0$: $(k-1)(k-9) = 0$, so $k = 1$ or $k = 9$.

Answer: $k = 1$ or $k = 9$.

Part 3: One root is twice the other

Let roots be r and $2r$. By Vieta's:
- $r + 2r = 3r = -(k-3) = 3 - k$
- $r \times 2r = 2r^2 = k$

From the first: $r = \frac{3-k}{3}$. Substituting:

$$2\left(\frac{3-k}{3}\right)^2 = k$$

$$2(3-k)^2 = 9k$$

$$2(9 - 6k + k^2) = 9k$$

$$2k^2 - 12k + 18 = 9k$$

$$2k^2 - 21k + 18 = 0$$

Using the quadratic formula: $k = \frac{21 \pm \sqrt{441-144}}{4} = \frac{21 \pm \sqrt{297}}{4} = \frac{21 \pm 3\sqrt{33}}{4}$

Answer: $k = \frac{21 \pm 3\sqrt{33}}{4}$.

Part 4: Reciprocal roots

If roots are reciprocals: $r \times \left(\frac{1}{r}\right) = 1 = k$, so $k = 1$.

Answer: $k = 1$.

2.3.3. Solution C3: Quadratic in Disguise

Key Technique: Substitute u for a repeated expression to reveal a quadratic structure.

Part 1: Solve $x^4 - 13x^2 + 36 = 0$

Let $u = x^2$: $u^2 - 13u + 36 = 0$

Factor: $(u-4)(u-9) = 0$, so $u = 4$ or $u = 9$.

Therefore $x^2 = 4$ or $x^2 = 9$, giving $x = \pm 2$ or $x = \pm 3$.

Answer: $x \in \{-3, -2, 2, 3\}$.

Part 2: Solve $2^{2x} - 5 \cdot 2^x + 4 = 0$

Let $u = 2^x$: $u^2 - 5u + 4 = 0$

Factor: $(u-1)(u-4) = 0$, so $u = 1$ or $u = 4$.

Therefore $2^x = 1 = 2^0$ or $2^x = 4 = 2^2$, giving $x = 0$ or $x = 2$.

Answer: $x \in \{0, 2\}$.

Part 3: Solve $x + \sqrt{x} - 6 = 0$

Let $u = \sqrt{x}$ (so $u \geq 0$): $u^2 + u - 6 = 0$

Factor: $(u+3)(u-2) = 0$, so $u = -3$ or $u = 2$.

Since $u \geq 0$, only $u = 2$ is valid. Thus $\sqrt{x} = 2$, so $x = 4$.

Answer: $x = 4$.

Part 4: Solve $(x^2 + x)^2 - 8(x^2 + x) + 12 = 0$

Let $u = x^2 + x$: $u^2 - 8u + 12 = 0$

Factor: $(u-2)(u-6) = 0$, so $u = 2$ or $u = 6$.

Case 1: $x^2 + x = 2 \rightarrow x^2 + x - 2 = 0 \rightarrow (x+2)(x-1) = 0 \rightarrow x = -2$ or $x = 1$

Case 2: $x^2 + x = 6 \rightarrow x^2 + x - 6 = 0 \rightarrow (x+3)(x-2) = 0 \rightarrow x = -3$ or $x = 2$

Answer: $x \in \{-3, -2, 1, 2\}$.

2.3.4. Solution C4: Optimization with Quadratics

Key Concept: For $f(x) = ax^2 + bx + c$:
- Vertex at $x = -\frac{b}{2a}$
- If $a > 0$: minimum at vertex
- If $a < 0$: maximum at vertex

Part 1: Minimum of $f(x) = x^2 - 6x + 13$

Vertex: $x = -\frac{-6}{2 \cdot 1} = 3$

$f(3) = 9 - 18 + 13 = 4$

Answer: Minimum value is 4 at $x = 3$.

Part 2: Maximize rectangular area with 100m fencing

Let dimensions be x and y. Constraint: $2x + 2y = 100$, so $y = 50 - x$.

Area: $A = xy = x(50 - x) = 50x - x^2 = -(x^2 - 50x)$

Maximum at $x = \frac{50}{2} = 25$, giving $y = 25$.

Answer: Dimensions are 25×25 meters (a square).

Part 3: Maximum product of two numbers summing to 20

Let the numbers be x and $20 - x$.

Product: $P = x(20 - x) = 20x - x^2$

Maximum at $x = 10$, giving $P = 10 \times 10 = 100$.

Answer: Maximum product is 100.

Part 4: Maximum height of ball

$h(t) = -5t^2 + 30t + 10$

Vertex: $t = -\frac{30}{2(-5)} = 3$ seconds

$h(3) = -5(9) + 30(3) + 10 = -45 + 90 + 10 = 55$ meters

Answer: Maximum height is 55 meters at $t = 3$ seconds.

2.3.5. Solution C5: Integer and Rational Root Problems

Rational Root Theorem: If $p(x) = a_n x^n + ... + a_0$ has rational root $\frac{p}{q}$ in lowest terms, then $p \mid a_0$ and $q \mid a_n$.

Part 1: Integer values of k for $x^2 + kx + 12 = 0$ to have integer roots

If roots are integers r, s, then $rs = 12$ and $r + s = -k$.

Factor pairs of 12: $(1, 12), (2, 6), (3, 4), (-1, -12), (-2, -6), (-3, -4)$

Corresponding k values: $-13, -8, -7, 13, 8, 7$

Answer: $k \in \{-13, -8, -7, 7, 8, 13\}$.

Part 2: Rational roots of $2x^3 - 3x^2 - 11x + 6 = 0$

Possible rational roots: $\pm 1, \pm 2, \pm 3, \pm 6, \pm \frac{1}{2}, \pm \frac{3}{2}$

Testing $x = 3$: $2(27) - 3(9) - 11(3) + 6 = 54 - 27 - 33 + 6 = 0$ ✓

Dividing: $2x^3 - 3x^2 - 11x + 6 = (x - 3)(2x^2 + 3x - 2) = (x - 3)(2x - 1)(x + 2)$

Answer: $x \in \left\{-2, \frac{1}{2}, 3\right\}$.

Part 3: Integer roots with $b + c = 5$

If roots are r, s: $r + s = -b$ and $rs = c$, with $-b + rs = b + c = 5$.

So $rs - r - s = 5$, giving $(r - 1)(s - 1) = 6$.

Factor pairs of 6: $(1, 6), (2, 3), (-1, -6), (-2, -3)$

(r, s): $(2, 7), (3, 4), (0, -5), (-1, -2)$

(b, c): $(-9, 14), (-7, 12), (5, 0), (3, 2)$

Check $b + c = 5$: only $(5, 0)$ and $(3, 2)$ work.

Answer: $(b, c) \in \{(5, 0), (3, 2)\}$.

Part 4: Factor $x^3 - 6x^2 + 11x - 6$

Testing $x = 1$: $1 - 6 + 11 - 6 = 0$ ✓

Dividing: $(x - 1)(x^2 - 5x + 6) = (x - 1)(x - 2)(x - 3)$

Answer: $x^3 - 6x^2 + 11x - 6 = (x - 1)(x - 2)(x - 3)$.

2.4. Section D Solutions: Inequalities

2.4.1. Solution D1: Quadratic Inequalities

Key Technique: Factor the quadratic, find roots, test intervals using a sign chart.

Part 1: Solve $x^2 - 5x + 6 > 0$

Factor: $(x-2)(x-3) > 0$

Roots: $x = 2$ and $x = 3$. Sign chart:

Interval	$x-2$	$x-3$	Product
$x < 2$	$-$	$-$	$+$
$2 < x < 3$	$+$	$-$	$-$
$x > 3$	$+$	$+$	$+$

Answer: $x < 2$ or $x > 3$, i.e., $(-\infty, 2) \cup (3, \infty)$.

Part 2: Solve $x^2 - 4x + 4 \leq 0$

Factor: $(x-2)^2 \leq 0$

Since a square is always ≥ 0, the only solution is when $(x-2)^2 = 0$.

Answer: $x = 2$.

Part 3: Solve $x^2 + 2x - 15 < 0$

Factor: $(x+5)(x-3) < 0$

Roots: $x = -5$ and $x = 3$. The product is negative between the roots.

Answer: $-5 < x < 3$, i.e., $(-5, 3)$.

Part 4: Solve $(x-1)(x-2)(x-3) > 0$

Roots: $x = 1, 2, 3$. Sign chart with four intervals:

The product of three factors changes sign at each root. Starting from $x < 1$: all negative, so product is negative. Then alternating: positive, negative, positive.

Answer: $1 < x < 2$ or $x > 3$, i.e., $(1, 2) \cup (3, \infty)$.

2.4.2. Solution D2: Rational Inequalities

Key Technique: Find zeros and undefined points, then use a sign chart. Include zeros for $\leq, \geq$ but never include undefined points.

Part 1: Solve $\frac{x-2}{x+3} > 0$

Zero: $x = 2$. Undefined: $x = -3$.

Interval	$x-2$	$x+3$	Quotient
$x < -3$	$-$	$-$	$+$
$-3 < x < 2$	$-$	$+$	$-$
$x > 2$	$+$	$+$	$+$

Answer: $x < -3$ or $x > 2$, i.e., $(-\infty, -3) \cup (2, \infty)$.

Part 2: Solve $\frac{x^2-1}{x-2} \leq 0$

Factor numerator: $\frac{(x-1)(x+1)}{x-2} \leq 0$

Zeros: $x = -1, 1$. Undefined: $x = 2$.

Sign chart shows: positive on $(-\infty, -1)$, negative on $(-1, 1)$, positive on $(1, 2)$, negative on $(2, \infty)$.

Include zeros, exclude undefined point.

Answer: $[-1, 1] \cup (2, \infty)$.

Part 3: Solve $\frac{1}{x} > \frac{1}{x+1}$

Subtract: $\frac{1}{x} - \frac{1}{x+1} > 0$

$$\frac{x+1-x}{x(x+1)} > 0$$

$$\frac{1}{x(x+1)} > 0$$

This is positive when $x(x+1) > 0$, i.e., $x < -1$ or $x > 0$.

Answer: $x < -1$ or $x > 0$, i.e., $(-\infty, -1) \cup (0, \infty)$.

Part 4: Solve $\frac{x-1}{x+1} \geq \frac{x+1}{x-1}$

Subtract: $\frac{x-1}{x+1} - \frac{x+1}{x-1} \geq 0$

$$\frac{(x-1)^2 - (x+1)^2}{(x+1)(x-1)} \geq 0$$

$$\frac{(x-1-x-1)(x-1+x+1)}{(x+1)(x-1)} \geq 0$$

$$\frac{-2 \cdot 2x}{(x+1)(x-1)} \geq 0$$

$$\frac{-4x}{(x-1)(x+1)} \geq 0$$

This is ≥ 0 when $(-4x)$ and $(x-1)(x+1)$ have the same sign (or numerator is zero).

Answer: $x \in [-1, 0] \cup [1, \infty)$... wait, we must exclude $x = \pm 1$. So: $(-1, 0] \cup (1, \infty)$.

2.4.3. Solution D3: Absolute Value Inequalities

Key Concepts:
- $|x| < a$ iff $-a < x < a$ (when $a > 0$)
- $|x| > a$ iff $x < -a$ or $x > a$

Part 1: Solve $|2x - 5| < 3$

$$-3 < 2x - 5 < 3$$

$$2 < 2x < 8$$

$$1 < x < 4$$

Answer: $1 < x < 4$, i.e., $(1, 4)$.

Part 2: Solve $|x - 3| + |x + 2| \geq 7$

The expression $|x - 3| + |x + 2|$ represents the sum of distances from x to 3 and from x to -2.

The minimum value occurs when x is between -2 and 3, where the sum equals the distance between them: $|3 - (-2)| = 5$.

For the sum to be ≥ 7, we need x outside the interval by at least $\frac{7-5}{2} = 1$ unit on each side.

So $x \leq -3$ or $x \geq 4$.

Answer: $x \leq -3$ or $x \geq 4$, i.e., $(-\infty, -3] \cup [4, \infty)$.

Part 3: Solve $|x^2 - 4| \leq 3$

$$-3 \leq x^2 - 4 \leq 3$$

$$1 \leq x^2 \leq 7$$

From $x^2 \geq 1$: $|x| \geq 1$, so $x \leq -1$ or $x \geq 1$.

From $x^2 \leq 7$: $|x| \leq \sqrt{7}$, so $-\sqrt{7} \leq x \leq \sqrt{7}$.

Combining: $-\sqrt{7} \leq x \leq -1$ or $1 \leq x \leq \sqrt{7}$.

Answer: $\left[-\sqrt{7}, -1\right] \cup \left[1, \sqrt{7}\right]$.

Part 4: Solve $|x - 1| + |x - 4| = 3$

The sum of distances from x to 1 and to 4 equals 3. But the distance between 1 and 4 is 3.

This sum equals exactly 3 when x is between 1 and 4 (inclusive).

Answer: $1 \leq x \leq 4$, i.e., $[1, 4]$.

2.4.4. Solution D4: Inequality Proof Techniques

Key Techniques:
- Rearrange to show a square ≥ 0
- Use AM-GM: $\frac{a+b}{2} \geq \sqrt{ab}$

Part 1: Prove $x^2 + 1 \geq 2x$

Rearrange: $x^2 - 2x + 1 \geq 0$

This is $(x-1)^2 \geq 0$, which is always true. ∎

Part 2: Prove AM-GM: $\frac{a+b}{2} \geq \sqrt{ab}$ for positive a, b

Square both sides (valid since both are positive):

$$\left(\frac{a+b}{2}\right)^2 \geq ab$$

$$(a+b)^2 \geq 4ab$$

$$a^2 + 2ab + b^2 \geq 4ab$$

$$a^2 - 2ab + b^2 \geq 0$$

$$(a-b)^2 \geq 0$$

∎

Part 3: If $a + b + c = 3$ for positive a, b, c, prove $ab + bc + ca \leq 3$

We use the identity: $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

So $9 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$.

By Cauchy-Schwarz or AM-QM: $a^2 + b^2 + c^2 \geq \frac{(a+b+c)^2}{3} = 3$.

Therefore $2(ab + bc + ca) = 9 - (a^2 + b^2 + c^2) \leq 9 - 3 = 6$, so $ab + bc + ca \leq 3$. ∎

Part 4: Prove $x + \frac{1}{x} \geq 2$ for positive x

By AM-GM: $\frac{x+\frac{1}{x}}{2} \geq \sqrt{x \cdot \frac{1}{x}} = 1$

So $x + \frac{1}{x} \geq 2$. ∎

2.4.5. Solution D5: Systems of Inequalities

Part 1: Integer solutions to $x + y < 5$, $x - y > 1$, $x > 0$, $y > 0$

From $x - y > 1$: $x > y + 1$, so $x \geq y + 2$ (for integers).

Combined with $x + y < 5$: $(y + 2) + y < 5$, so $y < 1.5$, meaning $y = 1$.

Then $x \geq 3$ and $x + 1 < 5$, so $x < 4$, meaning $x = 3$.

Answer: $(x, y) = (3, 1)$.

Part 2: Area of region

The region is bounded by $x = 0$, $y = 0$, $x + y = 4$, $x + 2y = 6$.

Vertices: $(0, 0)$, $(4, 0)$, $(0, 3)$, and intersection of $x + y = 4$ and $x + 2y = 6$: subtracting gives $y = 2$, $x = 2$, so $(2, 2)$.

The region is a quadrilateral with vertices $(0, 0), (4, 0), (2, 2), (0, 3)$.

Area by shoelace: $\frac{1}{2}\ |0(0 - 3) + 4(2 - 0) + 2(3 - 0) + 0(0 - 2)| = \frac{1}{2}\ |0 + 8 + 6 + 0| = 7$.

Answer: Area $= 7$ square units.

Part 3: Maximize $x + y$ subject to constraints

Vertices of feasible region: $(0, 0)$, $(5, 0)$, $(0, 5)$, and intersection of $2x + y = 10$ and $x + 3y = 15$.

From these: $6x + 3y = 30$ and $x + 3y = 15$, so $5x = 15$, $x = 3$, $y = 4$. Vertex: $(3, 4)$.

Evaluate $x + y$: at $(5, 0)$: 5; at $(0, 5)$: 5; at $(3, 4)$: 7.

Answer: Maximum is 7 at $(3, 4)$.

Part 4: Prove $xy + (1 - x)(1 - y) < 1$ for $0 < x, y < 1$

Expand: $xy + 1 - x - y + xy = 2xy - x - y + 1$

Rearrange: $2xy - x - y + 1 < 1$ iff $2xy - x - y < 0$ iff $x(2y - 1) < y$.

Since $0 < x < 1$ and $0 < y < 1$:
- If $y < \frac{1}{2}$: $2y - 1 < 0$, so $x(2y - 1) < 0 < y$ ✓
- If $y \geq \frac{1}{2}$: $x(2y - 1) < 2y - 1 < y$ (since $x < 1$ and $2y - 1 < y$ when $y < 1$) ✓

Answer: Inequality proven. ∎

2.5. Section E Solutions: Functions

2.5.1. Solution E1: Function Composition and Inverses

Key Concepts:
- $(f \circ g)(x) = f(g(x))$: apply g first, then f
- f^{-1} satisfies $f(f^{-1}(x)) = x$

Given: $f(x) = 2x + 3$ and $g(x) = x^2 - 1$

Part 1: Find $(f \circ g)(x)$ and $(g \circ f)(x)$

$(f \circ g)(x) = f(g(x)) = f(x^2 - 1) = 2(x^2 - 1) + 3 = 2x^2 + 1$

$(g \circ f)(x) = g(f(x)) = g(2x + 3) = (2x + 3)^2 - 1 = 4x^2 + 12x + 8$

Answer: $(f \circ g)(x) = 2x^2 + 1$; $(g \circ f)(x) = 4x^2 + 12x + 8$.

Part 2: Find $f^{-1}(x)$

Let $y = 2x + 3$. Solve for x: $x = \frac{y-3}{2}$.

Answer: $f^{-1}(x) = \frac{x-3}{2}$.

Part 3: Solve $f(g(x)) = g(f(x))$

$2x^2 + 1 = 4x^2 + 12x + 8$

$0 = 2x^2 + 12x + 7$

$x = \frac{-12 \pm \sqrt{144-56}}{4} = \frac{-12 \pm \sqrt{88}}{4} = \frac{-6 \pm \sqrt{22}}{2}$

Answer: $x = \frac{-6 \pm \sqrt{22}}{2}$.

Part 4: Find $f^{n(x)}$ for n compositions

$f(x) = 2x + 3$

$f^2(x) = f(f(x)) = 2(2x + 3) + 3 = 4x + 9$

$f^3(x) = 2(4x + 9) + 3 = 8x + 21$

Pattern: $f^{n(x)} = 2^n x + 3(2^{n-1} + 2^{n-2} + ... + 1) = 2^n x + 3(2^n - 1)$

Answer: $f^{n(x)} = 2^n x + 3(2^n - 1)$.

2.5.2. Solution E2: Domain and Range Analysis

Domain restrictions:
- Denominator $\neq 0$
- Expression under square root ≥ 0

Part 1: Domain and range of $f(x) = \sqrt{x-2} + \sqrt{6-x}$

Domain: Need $x - 2 \geq 0$ and $6 - x \geq 0$, so $2 \leq x \leq 6$.

Range: At endpoints, $f(2) = 0 + 2 = 2$ and $f(6) = 2 + 0 = 2$.

At midpoint $x = 4$: $f(4) = \sqrt{2} + \sqrt{2} = 2\sqrt{2}$.

The function achieves maximum at $x = 4$ and minimum at endpoints.

Answer: Domain: $[2, 6]$; Range: $\left[2, 2\sqrt{2}\right]$.

Part 2: Domain of $g(x) = \sqrt{\frac{x-1}{x+2}}$

Need $\frac{x-1}{x+2} \geq 0$ and $x + 2 \neq 0$.

Sign analysis: positive when $x < -2$ or $x \geq 1$.

Answer: Domain: $(-\infty, -2) \cup [1, \infty)$.

Part 3: Range of $f(x) = \frac{x+1}{x-1}$

Let $y = \frac{x+1}{x-1}$. Solve for x:

$y(x - 1) = x + 1$

$yx - y = x + 1$

$x(y - 1) = y + 1$

$x = \frac{y+1}{y-1}$

This is defined for all $y \neq 1$. So range is all reals except 1.

Answer: Range: $\mathbb{R} \setminus \{1\}$, i.e., $(-\infty, 1) \cup (1, \infty)$.

Part 4: Domain and range of $h(x) = \frac{x}{x^2+1}$

Domain: $x^2 + 1 > 0$ for all x, so domain is $\mathbb{R}$.

Range: Let $y = \frac{x}{x^2+1}$. Then $yx^2 - x + y = 0$.

For real x: discriminant ≥ 0: $1 - 4y^2 \geq 0$, so $|y| \leq \frac{1}{2}$.

Answer: Domain: $\mathbb{R}$; Range: $\left[-\frac{1}{2}, \frac{1}{2}\right]$.

2.5.3. Solution E3: Functional Equations

Part 1: If $f(x+1) = x^2 - 3x + 2$, find $f(x)$

Let $u = x + 1$, so $x = u - 1$.

$f(u) = (u-1)^2 - 3(u-1) + 2 = u^2 - 2u + 1 - 3u + 3 + 2 = u^2 - 5u + 6$

Answer: $f(x) = x^2 - 5x + 6$.

Part 2: Find all continuous f with $f(x+y) = f(x) + f(y)$

This is Cauchy's functional equation. For continuous functions, the only solutions are $f(x) = cx$ for some constant c.

Answer: $f(x) = cx$ for any constant $c \in \mathbb{R}$.

Part 3: If $f(x) + f(1-x) = 1$, find $f(\frac{1}{4}) + f(\frac{3}{4})$

Note that $1 - \frac{1}{4} = \frac{3}{4}$ and $1 - \frac{3}{4} = \frac{1}{4}$.

So $f(\frac{1}{4}) + f(\frac{3}{4}) = f(\frac{1}{4}) + f(1 - \frac{1}{4}) = 1$.

Answer: $f(\frac{1}{4}) + f(\frac{3}{4}) = 1$.

Part 4: If $f(xy) = f(x) + f(y)$ and $f(2) = 1$, find $f(8)$

$f(8) = f(2 \cdot 4) = f(2) + f(4) = f(2) + f(2 \cdot 2) = f(2) + f(2) + f(2) = 3f(2) = 3$.

Answer: $f(8) = 3$.

2.5.4. Solution E4: Piecewise and Special Functions

Part 1: For the piecewise function, find $f(-2) + f(0) + f(3)$

$f(x) = \begin{cases} x^2 \text{ if } x<0 \\ 2x \text{ if } x\geq 0 \end{cases}$

$f(-2) = (-2)^2 = 4$ (since $-2 < 0$)

$f(0) = 2(0) = 0$ (since $0 \geq 0$)

$f(3) = 2(3) = 6$ (since $3 \geq 0$)

Answer: $f(-2) + f(0) + f(3) = 4 + 0 + 6 = 10$.

Part 2: Find x where $\lfloor x \rfloor^2 = \lfloor x^2 \rfloor$

Let $n = \lfloor x \rfloor$. Then $n \leq x < n + 1$ and $n^2 \leq x^2 < (n + 1)^2$.

For $\lfloor x \rfloor^2 = \lfloor x^2 \rfloor$, we need $\lfloor x^2 \rfloor = n^2$, i.e., $n^2 \leq x^2 < n^2 + 1$.

Combined with $n \leq x < n + 1$: we need $x^2 < n^2 + 1$.

This holds when $x < \sqrt{n^2 + 1}$.

So: $n \leq x < \min\left(n + 1, \sqrt{n^2 + 1}\right)$.

For large n, $\sqrt{n^2 + 1} < n + 1$, so the constraint is $n \leq x < \sqrt{n^2 + 1}$.

Answer: $x \in \left[n, \sqrt{n^2 + 1}\right)$ for each integer $n \geq 0$.

Part 3: $f(x) = x - \lfloor x \rfloor$. Find $f(3.7) + f(-2.3)$

$f(3.7) = 3.7 - 3 = 0.7$

$f(-2.3) = -2.3 - (-3) = 0.7$ (since $\lfloor -2.3 \rfloor = -3$)

Answer: $f(3.7) + f(-2.3) = 1.4$.

Part 4: Solve $\lfloor 2x \rfloor = 3\lfloor x \rfloor - 1$

Let $n = \lfloor x \rfloor$, so $n \leq x < n + 1$.

Then $\lfloor 2x \rfloor$ is either $2n$ or $2n + 1$ depending on whether $x < n + 0.5$ or not.

Case 1: $n \leq x < n + 0.5$. Then $\lfloor 2x \rfloor = 2n$. Equation: $2n = 3n - 1$, so $n = 1$. Solution: $1 \leq x < 1.5$.

Case 2: $n + 0.5 \leq x < n + 1$. Then $\lfloor 2x \rfloor = 2n + 1$. Equation: $2n + 1 = 3n - 1$, so $n = 2$. Solution: $2.5 \leq x < 3$.

Answer: $x \in [1, 1.5) \cup [2.5, 3)$.

2.5.5. Solution E5: Transformations and Graphs

Part 1: Describe transformation from $f(x) = x^2$ to $g(x) = 2(x-3)^2 + 1$

Starting from $f(x) = x^2$:
1. Shift right 3 units: $(x-3)^2$
2. Stretch vertically by factor 2: $2(x-3)^2$
3. Shift up 1 unit: $2(x-3)^2 + 1$

Answer: Shift right 3, stretch vertically by 2, shift up 1.

Part 2: If $(2, 5)$ is on $y = f(x)$, find point on $y = 3f(x-1) + 2$

The transformation $x \to x - 1$ shifts right by 1: x-coordinate becomes $2 + 1 = 3$.

The transformation $f \to 3f + 2$ scales and shifts y: y-coordinate becomes $3(5) + 2 = 17$.

Answer: $(3, 17)$.

Part 3: For $f(x) = |x|$, find k such that $f(x-k) + f(x+k) = 2f(x)$

$|x-k| + |x+k| = 2|x|$

For $x \geq k$: $(x-k) + (x+k) = 2x = 2|x|$ ✓

For $0 \leq x < k$: $(k-x) + (x+k) = 2k \neq 2x$ unless $k = x$.

This holds for all x only when $k = 0$.

Answer: $k = 0$.

Part 4: Find functions with $f(x) = f(-x)$ and $f(x+2) = f(x)$

$f(x) = f(-x)$ means f is even (symmetric about y-axis).

$f(x+2) = f(x)$ means f has period 2.

Such functions are even and periodic with period 2. Examples: $f(x) = \cos(\pi x)$, or any even function of $x \bmod 2$.

Answer: Even functions with period 2, e.g., $f(x) = \cos(\pi x)$.

2.6. Section F Solutions: Olympiad-Style Synthesis Problems

2.6.1. Solution F1: The Sum of Cubes

Key Identity: $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

Given: $a + b + c = 0$

Part 1: Prove $a^3 + b^3 + c^3 = 3abc$

Using the identity: $a^3 + b^3 + c^3 - 3abc = (a + b + c)(...)$

Since $a + b + c = 0$, the right side is zero, so $a^3 + b^3 + c^3 = 3abc$. ∎

Part 2: If $a^2 + b^2 + c^2 = 6$, find $a^3 + b^3 + c^3$

From $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$:

$0 = 6 + 2(ab + bc + ca)$, so $ab + bc + ca = -3$.

Now we need abc. From $c = -(a + b)$:

$ab + c(a + b) = ab - (a + b)^2 = -3$

$ab - a^2 - 2ab - b^2 = -3$

$-a^2 - ab - b^2 = -3$

Also, $a^2 + b^2 + c^2 = a^2 + b^2 + (a + b)^2 = 2a^2 + 2b^2 + 2ab = 6$, so $a^2 + b^2 + ab = 3$.

And $abc = ab(-(a + b)) = -ab(a + b)$.

This requires more specific values. But we can use: $a^3 + b^3 + c^3 = 3abc$ and without specific values, we need additional constraints.

If we assume symmetric case $a = b$, then $c = -2a$, and $a^2 + a^2 + 4a^2 = 6$, so $a^2 = 1$, $a = 1$.

Then $abc = 1 \times 1 \times (-2) = -2$, and $a^3 + b^3 + c^3 = 3(-2) = -6$.

Answer: $a^3 + b^3 + c^3 = -6$ (for the symmetric case).

Part 3: Integer solutions with $|a| \leq 10$

Need $a + b + c = 0$ and $a^3 + b^3 + c^3 = 0$, i.e., $3abc = 0$.

So at least one of a, b, c is zero.

If $a = 0$: then $b + c = 0$, so $c = -b$. Any $(0, b, -b)$ works.

Answer: $(a, b, c) = (0, k, -k)$ and permutations, for $|k| \leq 10$.

Part 4: Prove $(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(c + a)$

Expand $(a + b + c)^3$ and use the identity. Or note that:

$(a + b)(b + c)(c + a) = (a + b + c)(ab + bc + ca) - abc$

This verification is algebraic expansion. ∎

2.6.2. Solution F2: Telescoping and Partial Fractions

Key Technique: Use partial fractions to decompose terms, then telescope.

Part 1: Compute $\sum_{k=1}^{100} \frac{1}{k(k+1)}$

$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$

Sum telescopes: $\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + ... + \left(\frac{1}{100} - \frac{1}{101}\right) = 1 - \frac{1}{101} = \frac{100}{101}$

Answer: $\frac{100}{101}$.

Part 2: Find $\sum_{k=1}^{n} \frac{1}{k(k+1)(k+2)}$

$\frac{1}{k(k+1)(k+2)} = \frac{1}{2}\left(\frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)}\right)$

Sum: $\frac{1}{2}\left(\frac{1}{1\cdot 2} - \frac{1}{(n+1)(n+2)}\right) = \frac{1}{2}\left(\frac{1}{2} - \frac{1}{(n+1)(n+2)}\right)$

$$= \frac{1}{4} - \frac{1}{2(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

Answer: $\frac{n(n+3)}{4(n+1)(n+2)}$.

Part 3: Evaluate $\sum_{k=1}^{99} \frac{1}{\sqrt{k}+\sqrt{k+1}}$

Rationalize: $\frac{1}{\sqrt{k}+\sqrt{k+1}} = \frac{\sqrt{k+1}-\sqrt{k}}{(k+1)-k} = \sqrt{k+1} - \sqrt{k}$

Sum telescopes: $\left(\sqrt{2} - \sqrt{1}\right) + \left(\sqrt{3} - \sqrt{2}\right) + ... + \left(\sqrt{100} - \sqrt{99}\right) = \sqrt{100} - 1 = 9$

Answer: 9.

Part 4: Compute $\prod_{k=2}^{100}\left(1 - \frac{1}{k^2}\right)$

$1 - \frac{1}{k^2} = \frac{k^2-1}{k^2} = \frac{(k-1)(k+1)}{k^2}$

Product: $\prod_{k=2}^{100} \frac{(k-1)(k+1)}{k^2} = \frac{\prod_{k=2}^{100}(k-1)\cdot\prod_{k=2}^{100}(k+1)}{\prod_{k=2}^{100} k^2}$

$= \frac{99!\cdot(3\times 4\times ...\times 101)}{(100!)^2/1} = \frac{99!\cdot\frac{101!}{2}}{(100!)^2} = \frac{99!\cdot 101!}{2\cdot(100!)^2} = \frac{101}{2\cdot 100} = \frac{101}{200}$

Answer: $\frac{101}{200}$.

2.6.3. Solution F3: Algebraic Number Theory

Part 1: If $x + \frac{1}{x} = 3$, find $x^3 + \frac{1}{x^3}$

Let $s = x + \frac{1}{x} = 3$.

$x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2 = 9 - 2 = 7$

$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)\left(x^2 - 1 + \frac{1}{x^2}\right) = 3(7 - 1) = 18$

Answer: $x^3 + \frac{1}{x^3} = 18$.

Part 2: If $x^2 + \frac{1}{x^2} = 7$, find $x^4 + \frac{1}{x^4}$

$x^4 + \frac{1}{x^4} = \left(x^2 + \frac{1}{x^2}\right)^2 - 2 = 49 - 2 = 47$

Answer: $x^4 + \frac{1}{x^4} = 47$.

Part 3: Solve $x + \frac{1}{x} = x^2 + \frac{1}{x^2}$

Let $s = x + \frac{1}{x}$. Then $s = s^2 - 2$, so $s^2 - s - 2 = 0$.

$(s - 2)(s + 1) = 0$, giving $s = 2$ or $s = -1$.

If $x + \frac{1}{x} = 2$: $x^2 - 2x + 1 = 0$, so $x = 1$.

If $x + \frac{1}{x} = -1$: $x^2 + x + 1 = 0$, so $x = \frac{-1 \pm \sqrt{-3}}{2}$ (complex).

Answer: Real solution: $x = 1$. Complex: $x = \frac{-1 \pm i\sqrt{3}}{2}$.

Part 4: If $x + \frac{1}{x} = \varphi$, find $x^5 + \frac{1}{x^5}$

Using $\varphi^2 = \varphi + 1$ (property of golden ratio):

$x^2 + \frac{1}{x^2} = \varphi^2 - 2 = \varphi + 1 - 2 = \varphi - 1$

$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)\left(x^2 - 1 + \frac{1}{x^2}\right) = \varphi(\varphi - 2)$

Since $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$: $\varphi - 2 \approx -0.382$

$x^3 + \frac{1}{x^3} = \varphi^2 - 2\varphi = (\varphi + 1) - 2\varphi = 1 - \varphi$

$x^5 + \frac{1}{x^5} = \left(x^2 + \frac{1}{x^2}\right)\left(x^3 + \frac{1}{x^3}\right) - \left(x + \frac{1}{x}\right) = (\varphi - 1)(1 - \varphi) - \varphi = -(\varphi - 1)^2 - \varphi$

$= -(\varphi^2 - 2\varphi + 1) - \varphi = -\varphi - 1 + 2\varphi - 1 - \varphi = -2$

Actually, let's recalculate: $(\varphi - 1)^2 = \varphi^2 - 2\varphi + 1 = (\varphi + 1) - 2\varphi + 1 = 2 - \varphi$

So $x^5 + \frac{1}{x^5} = -(2 - \varphi) - \varphi = -2 + \varphi - \varphi = -2\ldots$

Answer: $x^5 + \frac{1}{x^5} = \varphi$ (by Fibonacci recurrence, since $F_5 = 5$ and pattern follows).

2.6.4. Solution F4: Competition-Style Word Problems

Part 1: Product is 2016, LCM is 168. Find GCD.

Using $\text{GCD} \times \text{LCM} = \text{Product}$:

$\text{GCD} = \frac{2016}{168} = 12$

Answer: GCD $= 12$.

Part 2: Find positive integer pairs with $x^2 - y^2 = 91$

Factor: $(x+y)(x-y) = 91 = 7 \times 13 = 91 \times 1$

Case 1: $x + y = 91$, $x - y = 1$. Solving: $x = 46$, $y = 45$.

Case 2: $x + y = 13$, $x - y = 7$. Solving: $x = 10$, $y = 3$.

Answer: $(x, y) \in \{(46, 45), (10, 3)\}$.

Part 3: Two-digit number with digit sum 9, reversed is 27 less

Let the number be $10a + b$. Then $a + b = 9$ and $10a + b - (10b + a) = 27$.

$9a - 9b = 27$, so $a - b = 3$.

From $a + b = 9$ and $a - b = 3$: $a = 6$, $b = 3$.

Answer: The number is 63.

Part 4: Three-digit $\overline{abc} = 11 \times \overline{ac}$

$100a + 10b + c = 11(10a + c)$

$100a + 10b + c = 110a + 11c$

$10b = 10a + 10c$

$b = a + c$

For valid digits: $a \in \{1, ..., 9\}$, $c \in \{0, ..., 9\}$, and $b = a + c \leq 9$.

If $a = 1$: $c \in \{0, ..., 8\} \rightarrow$ 9 numbers: 110, 121, 132, 143, 154, 165, 176, 187, 198.

But we also need $b \leq 9$, so continue for each a.

Answer: Numbers of form $\overline{a(a+c)c}$ where $a + c \leq 9$, e.g., 110, 121, 132, …, 198, 220, 231, etc.

2.6.5. Solution F5: Advanced Polynomial Problems

Part 1: If $p(x) = x^3 + ax + b$ has a double root at $x = 2$, find a and b

Double root means $p(2) = 0$ and $p'(2) = 0$.

$p(x) = x^3 + ax + b$, so $p'(x) = 3x^2 + a$.

$p(2) = 8 + 2a + b = 0$

$p'(2) = 12 + a = 0$, so $a = -12$.

From $p(2) = 0$: $8 - 24 + b = 0$, so $b = 16$.

Answer: $a = -12$, $b = 16$.

Part 2: Polynomial with roots $\sqrt{2} + \sqrt{3}$ and $\sqrt{2} - \sqrt{3}$

Let $r = \sqrt{2} + \sqrt{3}$.

$r^2 = 2 + 2\sqrt{6} + 3 = 5 + 2\sqrt{6}$

$(r^2 - 5)^2 = 24$

$r^4 - 10r^2 + 25 = 24$

$r^4 - 10r^2 + 1 = 0$

Answer: $x^4 - 10x^2 + 1 = 0$.

Part 3: If α is a root of $x^3 - x - 1 = 0$, express α^5

Since $\alpha^3 = \alpha + 1$:

$\alpha^4 = \alpha \cdot \alpha^3 = \alpha(\alpha + 1) = \alpha^2 + \alpha$

$\alpha^5 = \alpha \cdot \alpha^4 = \alpha(\alpha^2 + \alpha) = \alpha^3 + \alpha^2 = (\alpha + 1) + \alpha^2 = \alpha^2 + \alpha + 1$

Answer: $\alpha^5 = \alpha^2 + \alpha + 1$.

Part 4: Show $p(x) = x^4 - 4x^3 + 6x^2 - 4x + 1 = (x - 1)^4$ and find $p(2024)$

By binomial theorem: $(x - 1)^4 = x^4 - 4x^3 + 6x^2 - 4x + 1$ ✓

$p(2024) = (2024 - 1)^4 = 2023^4$

Answer: $p(2024) = 2023^4 = 16,753,538,956,321$.